YOUNGKI YOU

Marysville, WA youngki.you.research@gmail.com https://youngkiyou.github.io

PROFESSIONAL SUMMARY

Structural biologist and proteomics researcher investigating how molecular interactions, conformational changes, and post-translational modifications regulate protein function across molecular and cellular scales. My research integrates mechanistic structural biology, structural proteomics, quantitative mass spectrometry, protein biochemistry, cellular assays, and computational modeling to connect residue-level perturbations with protein assemblies, interaction networks, and functional outcomes. My work has included membrane-associated and neuronal signaling proteins, large-scale protein interaction mapping, and PTM-dependent regulation in inflammatory and disease-relevant biological systems. Across these projects, I have focused on defining how changes in protein structure, molecular interactions, and chemical modification alter regulatory behavior and cellular function. I am particularly interested in developing integrative approaches that bridge structural mechanisms with protein interaction networks and cellular regulation.

RESEARCH EXPERTISE

Molecular Recognition & Structural Regulation

- Structural and biochemical analysis of protein–ligand and protein–protein interactions in regulatory protein systems
- Ligand-dependent conformational regulation, membrane-associated signaling, and structure–function relationships
- X-ray crystallography, NMR sample preparation, and integrative structural interpretation
- Protein engineering, construct design, mutagenesis, and stabilization for mechanistic studies
- Structure-guided analysis of interaction interfaces, dynamic protein states, and regulatory mechanisms

Protein Interaction Architecture & Structural Proteomics

- XL-MS and LC-MS/MS workflows for residue-level structural restraints, protein interaction architecture, and complex organization
- Proteome-scale XL-MS interaction mapping to identify interaction hubs and infer functional organization
- Structural interpretation of apolipoprotein, lipoprotein, biotin carboxylase, and microbial interaction systems
- Integration of XL-MS, biochemical, cryo-EM, and computational data to generate mechanistic models of protein assemblies
- XL-MS data analysis using XlinkX and pLink
- Nanobody-based binder development for probing interaction specificity and complex organization

Post-Translational Regulation & Functional Proteomics

- LC-MS/MS-based proteomics for analysis of protein modifications and regulatory mechanisms
- PTM analysis including carbamylation, acetylation, phosphorylation, and methylation
- PTM crosstalk and modification-dependent regulation linking protein chemistry with cellular function and disease-relevant phenotypes
- Quantitative proteomics and data analysis using FragPipe, MaxQuant, and Perseus
- Experience with Thermo QE-HF, Lumos Tribrid, and Eclipse Tribrid mass spectrometry platforms

Computational Structural Biology & Data Integration

- Molecular docking and molecular dynamics simulations for characterization of protein–ligand, protein–membrane, and protein–protein interactions
- Structure prediction and modeling using AlphaFold and RoseTTAFold
- Structural visualization and analysis using PyMOL, Chimera, and Discovery Studio
- Python-, R-, and Linux-based workflows for quantitative analysis and integration of structural and proteomic data

Protein Biochemistry & Cellular Characterization

- Protein expression, purification, and stabilization for structural and functional studies, including affinity chromatography, ion-exchange chromatography, SEC (ÄKTA), and isotopic labeling
- Biochemical and biophysical characterization of protein interactions and regulatory mechanisms
- Mammalian and bacterial cell culture, transformation, and transfection
- Immunoprecipitation, Western blotting, ELISA, confocal microscopy, and cell-based functional assays
- Experimental troubleshooting, workflow optimization, project leadership, and cross-disciplinary collaboration

RESEARCH EXPERIENCE

Pacific Northwest National Laboratory, Biological Sciences Division Richland, WA

Postdoctoral Research Associate *02/2022 – 02/2026*

Research Focus: Structural proteomics and PTM biology to define protein interaction architecture, conformational regulation, and modification-dependent function across molecular and cellular systems.

- **Protein Interaction Architecture & Structural Proteomics**
  - Mapped protein interaction architectures in apolipoprotein and lipoprotein systems using XL-MS, biochemical analysis, and structural modeling to define residue-level contacts and organization of lipid-associated protein assemblies.
  - Applied XL-MS-derived structural restraints and computational modeling to interpret protein interfaces and higher-order assemblies across apolipoprotein and other protein systems.
  - Investigated oligomeric organization of plant biotin carboxylase through integration of XL-MS with cryo-EM and structural analysis, supporting models of higher-order assembly.
  - Developed and characterized nanobody-based binders to probe interaction specificity and support structural and biochemical analysis of lipoprotein complexes.
- **Proteome-Scale Interaction Networks**
  - Developed and applied XL-MS workflows for large-scale protein interaction mapping, generating approximately 2,700 inter-protein cross-links in cyanobacterial systems.
  - Mapped proteome-scale interaction networks in *Chlamydomonas reinhardtii* and *Prochlorococcus* MED4 to identify interaction hubs and infer functional organization across cellular pathways.
  - Established reproducible experimental and analytical workflows for complex structural-proteomics datasets and integrated interaction data with structural and computational models.
- **Post-Translational Regulation & Functional Proteomics**
  - Developed LC-MS/MS workflows for carbamylation and multi-PTM analysis, identifying more than 8,000 carbamylated and more than 7,000 acetylated peptides and expanding coverage of modification-dependent regulation.

- Investigated PTM-dependent regulation in inflammatory and type 1 diabetes-related systems, including carbamylation, PTM crosstalk, and ARH3-associated regulation of inflammatory signaling.
- Combined quantitative proteomics with pathway, structural, and functional analyses to connect PTM changes with molecular and cellular phenotypes.
- Investigated evolutionarily conserved bacterial protein methylation and its association with stringent-response signaling and ribosomal regulation.

- **Data Integration & Mechanistic Interpretation**
  - Developed PTMOverlay, a computational tool for visualization and interpretation of complex PTM datasets across biological systems.
  - Integrated structural, biochemical, proteomic, and computational datasets to generate mechanistic models of protein interactions, regulatory assemblies, and modification-dependent protein function.
  - Collaborated across multidisciplinary teams to connect unbiased proteomic discovery with targeted biochemical, structural, and functional validation.

University of Houston College of Pharmacy Houston, TX

Postdoctoral Fellow *01/2021 – 07/2021*

Graduate Research Assistant *08/2015 – 12/2020*

Research Focus: Mechanistic structural analysis of molecular recognition and conformational regulation in the presynaptic signaling protein Munc13-1 and related membrane-associated protein systems.

- **Munc13-1 Structural Regulation & Membrane Signaling**
  - Defined structural determinants of ligand recognition and membrane interaction in the Munc13-1 C1 domain through X-ray crystallography, protein biochemistry, mutagenesis, molecular docking, and molecular dynamics simulations.
  - Characterized apo and ligand-bound structural states of Munc13-1 C1 and identified residues, including Trp-588, that contribute to ligand binding, membrane association, and structure–function relationships.
  - Engineered and characterized wild-type and mutant proteins to test how residue-level structural perturbations alter ligand-dependent behavior.
  - Expressed and purified unlabeled and ^{15}N-labeled Munc13-1 C1 proteins and optimized constructs, buffer conditions, stability, and yield for structural and biophysical studies.
  - Used cell-based assays and confocal microscopy to examine ligand-dependent membrane recruitment and intracellular localization of Munc13-1, connecting molecular recognition with cellular behavior.
- **Computational Structural Biology & Collaborative Modeling**
  - Applied molecular docking and molecular dynamics simulations to characterize protein–ligand and membrane-associated interactions and interpret conformational changes associated with ligand binding.
  - Extended structure-based analysis to additional regulatory systems, including PKC domains, proteasome-associated targets, ACE2, and TET2–DNA complexes.
  - Developed standardized molecular dynamics workflows that improved reproducibility and supported collaborative mechanistic studies.

Handong Institute of Life Science, Handong Global University Pohang, South Korea
Graduate Research Assistant *02/2010 – 09/2014*
Research Assistant *12/2006 – 02/2007; 07/2007 – 01/2010*

Research Focus: Computational analysis of protein sequence–structure relationships and structure-based functional modeling.

- Developed an amino-acid substitution matrix using structural and physicochemical properties of protein blocks through PCA-based computational analysis.
- Investigated relationships among protein sequence, structural organization, and evolutionary conservation.
- Applied computational structural analysis to biological problems including antiplatelet compounds and intranasal vaccine candidates.

EDUCATION

Ph.D., Pharmacology and Medicinal Chemistry - University of Houston**,** Houston, TX, 2020
M.Sc., Life Science - Graduate School, Handong Global University**,** Pohang, South Korea, 2012
B.Sc., Life Science (minor in Computer Science) - Handong Global University, Pohang, South Korea, 2010

PUBLICATIONS

24 peer-reviewed publications, including 10 first- or co-first-author papers

Peer-Reviewed Publications

† Equal contribution.

1. Madison, H.J., Dunn, L., **You, Y.**, Lemes Jorge, G., Paša-Tolić, L., Thelen, J.J., Van Doren, S.R., and Yokom, A.L. "Oligomeric assemblies of plant biotin carboxylase revealed by cryo-EM and cross-linking." *Biochemical Journal* (2026): BCJ20250372.
2. He, Y., Song, H., **You, Y.**, et al. "A model of human APOA2 on HDL." *Journal of Lipid Research* (2026): 67(9), 101113.
3. Powell, S.M., **You, Y.**, Ogbu, C., and Melchior, J.T. "The structural and functional journey of Apolipoprotein A-I through the human body." *Current Atherosclerosis Reports* (2026): 38.
4. Neupane, N.P.†, Mathews, J.T.†, Lingaraju, S.†, **You, Y.†**, Mathukumalli, K., and Das, J. "Mitragynine and morphine stabilize different conformations of the μ-opioid receptor as revealed by molecular dynamics simulation studies." *ACS Omega* (2026).
5. **You, Y.**, Sarkar, S., Deiter, C., Elliott, E., Nicora, C., Mirmira, R., Sussel, L., and Nakayasu, E.S. "Reduction of chemokine CXCL9 expression by omega-3 fatty acids via ADP-ribosylhydrolase ARH3 in MIN6 insulin-producing cells." *Proteomics* (2025): e202400053.
6. **You, Y.**, Many, G., and Nakayasu, E.S. "Protein carbamylation and proteomics: from artifacts to elucidation of biological functions." *Frontiers in Analytical Science* (2025): 1512573.
7. Dakup, P.P., Diaz Ludovico, I., **You, Y.**, Rao, C., Flores, J., Bramer, L.M., Rewers, M., Webb-Robertson, B.J.M., Metz, T.O., Mirmira, R.G., Sims, E.K., and Nakayasu, E.S. "Challenges and opportunities in state-of-the-art proteomics analysis for biomarker development from plasma extracellular vesicles." *Proteomics* (2025): e70036.
8. Sarkar, S., Morris, J., **You, Y.**, Sexmith, H., Street, S.E., Thibert, S.M., Attah, I.K., Hutchinson Bunch, C.M.,

Novikova, I.V., Evans, J.E., Shah, A.S., Gordon, S.M., Segrest, J.P., Bornfeldt, K.E., Vaisar, T., Heinecke, J.W., Davidson, W.S., and Melchior, J.T. "APOA2 increases cholesterol efflux capacity to plasma HDL by displacing the C-terminus of resident APOA1." *Journal of Lipid Research* (2024): 65(12), 100686.

9. Sarkar, S., Zheng, X., Clair, G., Kwon, Y., **You, Y.**, Swensen, A.C., Webb-Robertson, B.J.M., Nakayasu, E.S., Qian, W.-J., and Metz, T.O. "Exploring new frontiers in type 1 diabetes through advanced mass spectrometry-based molecular measurements." *Trends in Molecular Medicine* (2024).
10. Walukiewicz, H., Farris, Y., Burnet, M., Feid, S., **You, Y.**, Kim, H., Bank, T., Christensen, D., Payne, S.H., Wolfe, A., Rao, C., and Nakayasu, E.S. "Regulation of bacterial stringent response by an evolutionarily conserved ribosomal protein L11 methylation." *mBio* (2024): e01773-24.
11. **You, Y.**[†], Tsai, C.-F.[†], Patel, R., Sarkar, S., Clair, G., Zhou, M., Liu, T., Metz, T.O., Das, C., and Nakayasu, E.S. "Analysis of a macrophage carbamylated proteome reveals a function in post-translational modification crosstalk." *Cell Communication and Signaling* (2023): 21, 241.
12. **You, Y.**, Mathukumalli, K., and Das, J. "Comparison of the ligand binding site of C1 domains: a molecular dynamics simulation study of the C1 domain–phorbol 13-acetate–membrane system." *Journal of Biomolecular Structure and Dynamics* (2023): 11796–11809.
13. Vallejo, M.C., Sarkar, S., Elliott, E.C., Henry, H.R., Powell, S.M., Diaz Ludovico, I., **You, Y.**, Huang, F., Payne, S.H., Ramanadham, S., Sims, E.K., Metz, T.O., Mirmira, R.G., and Nakayasu, E.S. "A proteomic meta-analysis refinement of plasma extracellular vesicles." *Scientific Data* (2023): 837.
14. Das, J.[†], **You, Y.**[†], Mathukumalli, K., Ann, J., Lee, J., and Marquez, V.E. "Activation of Munc13-1 by Diacylglycerol (DAG)-Lactones." *Biochemistry* (2023): 2717–2726.
15. **You, Y.**, and Das, J. "Molecular dynamics simulation studies on binding of activator and inhibitor to Munc13-1 C1 in the presence of membrane." *Journal of Biomolecular Structure and Dynamics* (2022): 14160–14175.
16. **You, Y.**, Katti, S., Yu, B., Igumenova, T.I., and Das, J. "Probing the diacylglycerol binding site of presynaptic Munc13-1." *Biochemistry* (2021): 60(16), 1286–1298.
17. Khan, T.K., **You, Y.**, Nelson, T.J., Kundu, S., Pramanik, S., and Das, J. "Modulation of proteasome activity by curcumin and didemethylcurcumin." *Journal of Biomolecular Structure and Dynamics* (2021): 8332–8339.
18. **You, Y.**, and Das, J. "Effect of ethanol on Munc13-1 C1 in membrane: a molecular dynamics simulation study." *Alcoholism: Clinical and Experimental Research* (2020): 44(7), 1344–1355.
19. Blanco, F.A., Czikora, A., Kedei, N., **You, Y.**, Mitchell, G.A., Pany, S., Ghosh, A., Blumberg, P.M., and Das, J. "Munc13 is a molecular target of bryostatin 1." *Biochemistry* (2019): 58(27), 3016–3030.
20. Czikora, A., Pany, S., **You, Y.**, Saini, A.S., Lewin, N.E., Mitchell, G.A., Abramovitz, A., Kedei, N., Blumberg, P.M., and Das, J. "Structural determinants of phorbol ester binding activity of the C1a and C1b domains of protein kinase C theta." *Biochimica et Biophysica Acta - Biomembranes* (2018): 1860(5), 1046–1056.
21. Das, J., Kedei, N., Kelsey, J.S., **You, Y.**, Pany, S., Mitchell, G.A., Lewin, N.E., and Blumberg, P.M. "Critical role of Trp-588 of presynaptic Munc13-1 for ligand binding and membrane translocation." *Biochemistry* (2018): 57(5), 732–741.
22. Pany, S., Ghosh, A., **You, Y.**, Nguyen, N., and Das, J. "Resveratrol inhibits phorbol ester-induced membrane translocation of presynaptic Munc13-1." *Biochimica et Biophysica Acta - General Subjects* (2017): 1861(11), 2640–2651.
23. Pany, S., **You, Y.**, and Das, J. "Curcumin inhibits protein kinase Cα activity by binding to its C1 domain." *Biochemistry* (2016): 55(45), 6327–6336.

24. **You, Y.**, Jang, I.H., Lee, K., Kim, H.J., and Lee, K.H. "An approach for a substitution matrix based on protein blocks and physicochemical properties of amino acids through PCA." *Interdisciplinary Bio Central* (2014): 6(1), 3.

Manuscripts and Preprints

25. **You, Y**., Kim, H., Ushakumary, M.G., et al. "Carbamylome analysis reveals regulatory roles for lysine carbamylation in β-cell glycolytic and insulin-processing enzymes." *Biochemical Journal* (under revision).
26. Sarkar, S., Li, F., **You, Y.**, Elliott, E.C., Zemaitis, K.J., Kim, H., Markillie, L.M., Traeger, J., Powell, S.M., *et al.* "Insulin post-transcriptional regulation via PARP12-mediated ADP-ribosylation." *Cell Communication and Signaling* (under revision).
27. Krieger, C.[†], Everton, Z.[†], **You, Y.**[†], Lewis, B., Bank, T., Burnet, M.C., *et al.* "PTMOverlay: A proteomic tool to visualize post-translational modifications across evolution." *Proteomics* (under review).
28. Sarkar, S., Flores, J.E., Nune, N., Ramos-Rodriguez, M., *et al.*, **You, Y.**, and Nakayasu, E.S. "Multi-omics data compendium of pancreatic islet and β-cell responses to pro-inflammatory cytokines." *Scientific Data* (under review).
29. Van Fossen, E., Stephenson, M., Frank, A., Akella, S., Wooldridge, R., Wilson, A., **You, Y.**, Nakayasu, E.S., Egbert, R., and Elmore, J. "Efficient genetic code expansion tools enable in vivo study of lysine acetylation in non-model bacteria." *bioRxiv* (2025).

AWARDS

- 1st Place, Graduate Student Oral Presentation, University of Houston College of Pharmacy, 2019
- Travel Award, Biophysical Society, 2019